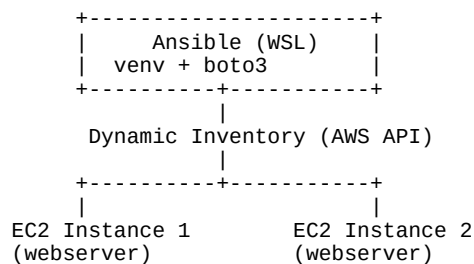


AWS + Ansible Dynamic Inventory Lab (Portfolio Version)

1. Overview

This lab demonstrates how to dynamically manage AWS EC2 instances using Ansible and tag-based discovery.

2. Architecture



3. Prerequisites

- AWS CLI configured (aws configure)
- Python3 + venv
- Ansible installed in venv
- boto3 / botocore installed

4. Environment Setup

```
python3 -m venv venv
source venv/bin/activate
pip install ansible boto3 botocore
```

5. Dynamic Inventory Configuration

```
plugin: amazon.aws.aws_ec2

regions:
  - ca-central-1

filters:
  instance-state-name: running
  "tag:Project": "ansible-lab"

compose:
  ansible_host: public_ip_address
```

6. AWS Tagging Strategy

Instances must include the following tags:

```
Key: Project
Value: ansible-lab
```

Key: Name
Value: webserver

7. Validation Commands

```
ansible-inventory --graph  
ansible-inventory --list
```

Expected Output

```
@aws_ec2:  
|--54.xxx.xxx.xxx  
|--52.xxx.xxx.xxx
```

8. Connectivity Test

```
ansible all -m ping -u ubuntu --private-key ~/.ssh/key.pem
```

9. Troubleshooting (Real Issues Encountered)

1. YAML errors (duplicate keys)
2. Wrong field name (public-ip vs public_ip)
3. venv not activated
4. boto3 missing
5. Tag filter mismatch
6. SSH permission denied

10. Lessons Learned

- Dynamic inventory relies entirely on AWS tags - Environment consistency (venv vs system Python) is critical - Debugging is a core DevOps skill - Always validate assumptions step by step

11. Extension (Terraform Integration)

```
resource "aws_instance" "web" {  
  count = 2  
  instance_type = "t3.micro"  
  
  tags = {  
    Name = "webserver"  
    Project = "ansible-lab"  
  }  
}
```